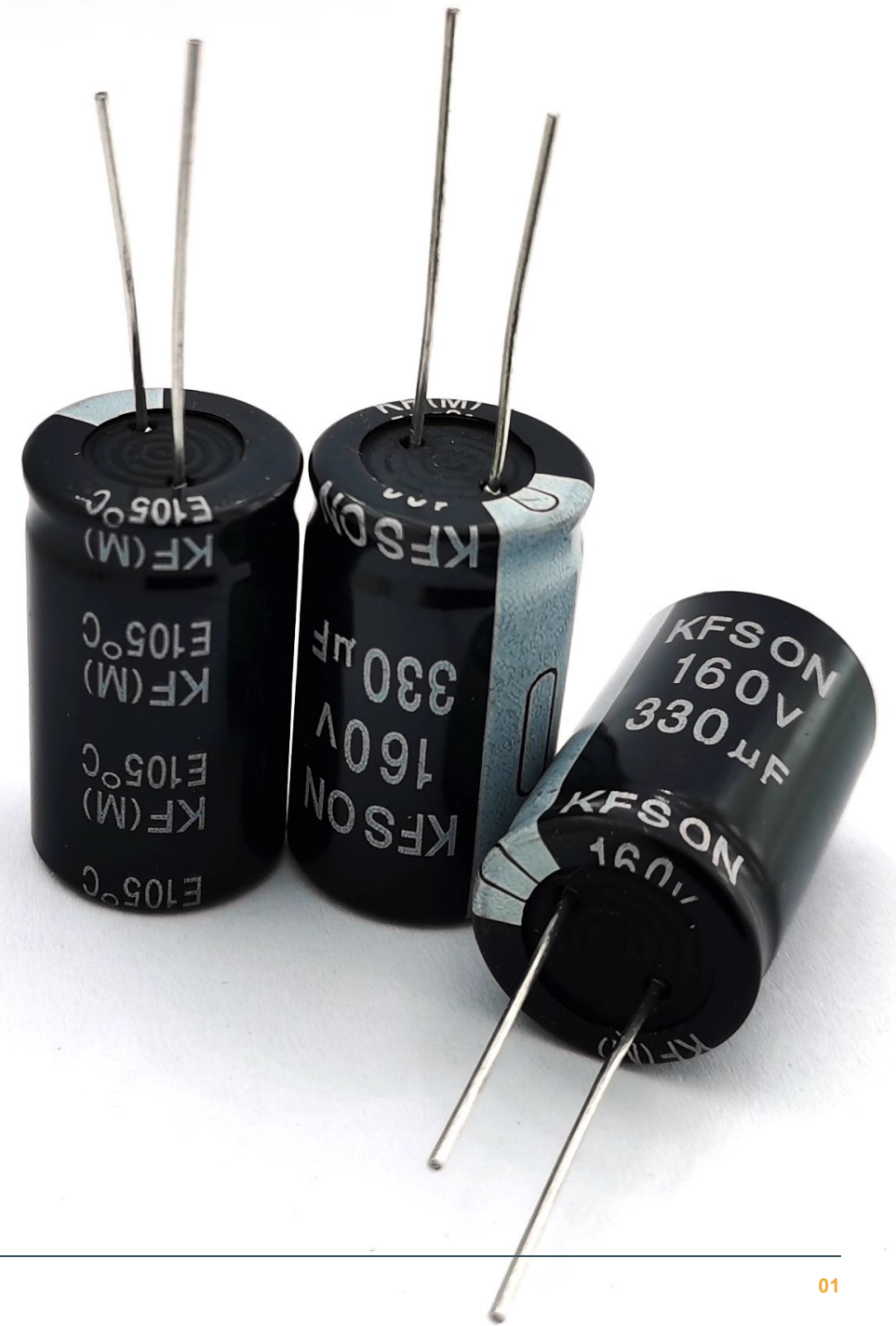


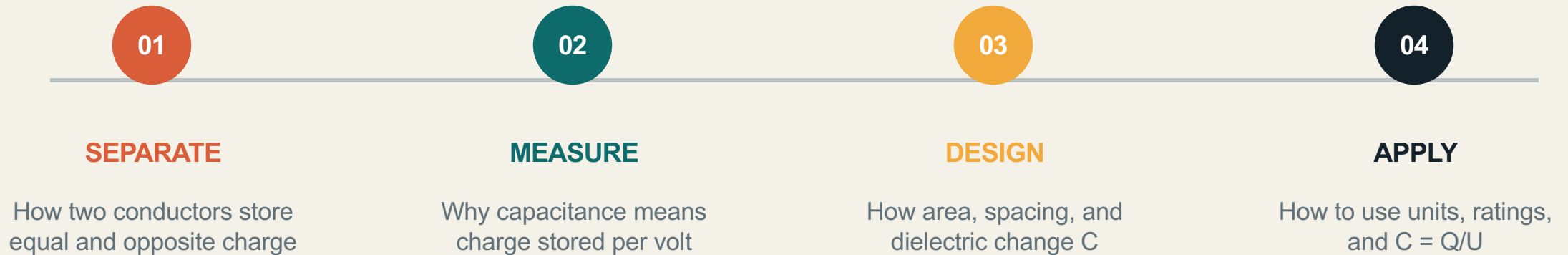
How much charge can one volt hold?

A lesson on capacitors, charge separation, and the meaning of $C = Q/U$

STORE • MEASURE • DESIGN



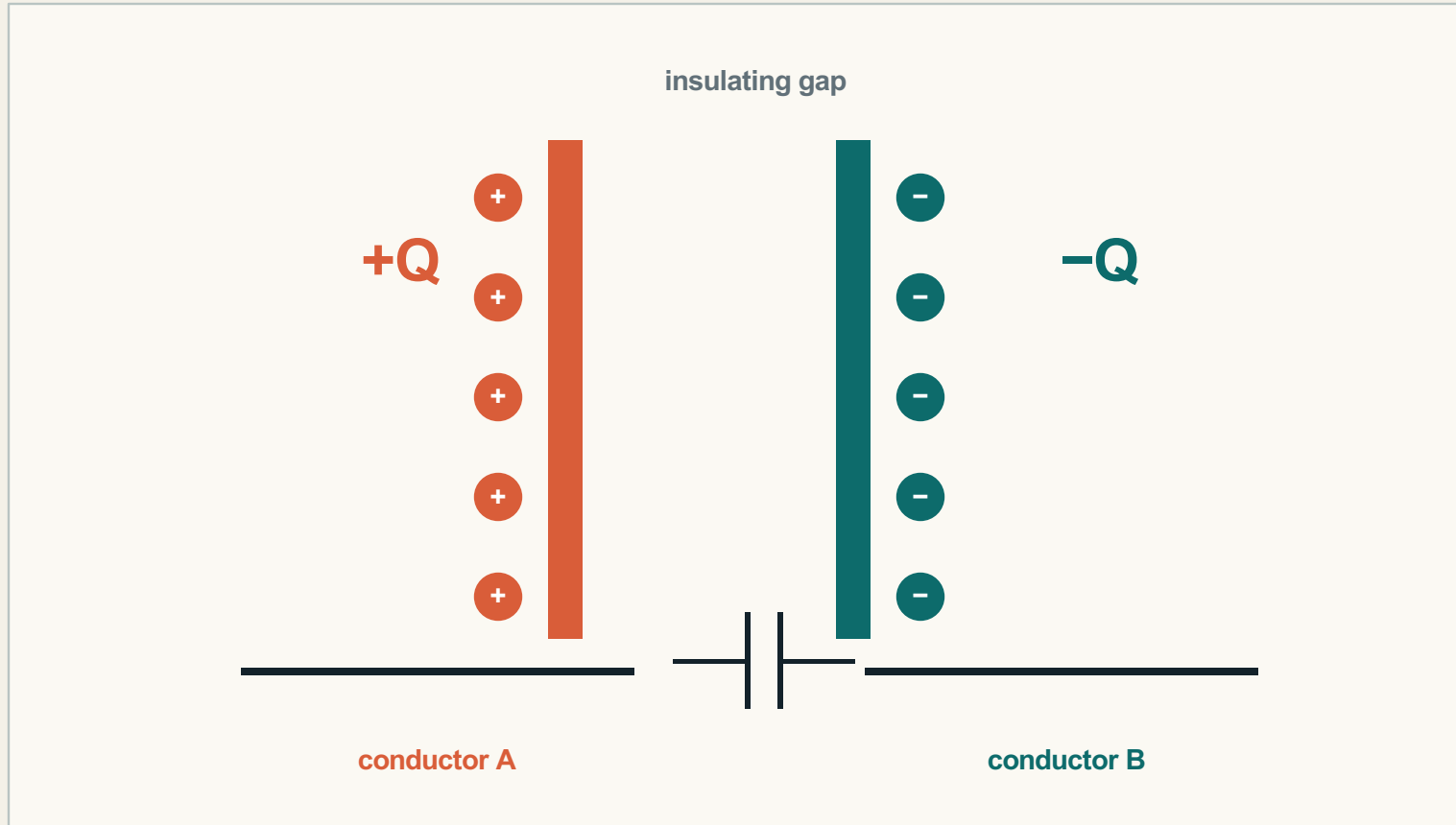
Build the idea in four moves.



Essential question

What does a capacitor's capacitance tell us?

A capacitor stores charge by separating it.



No net charge is created.

Electrons are moved from one conductor to the other. The plates carry equal-magnitude, opposite charges.

A capacitor needs two conductors separated by an insulator.

From Leyden jars to modern components, the structure is unchanged.



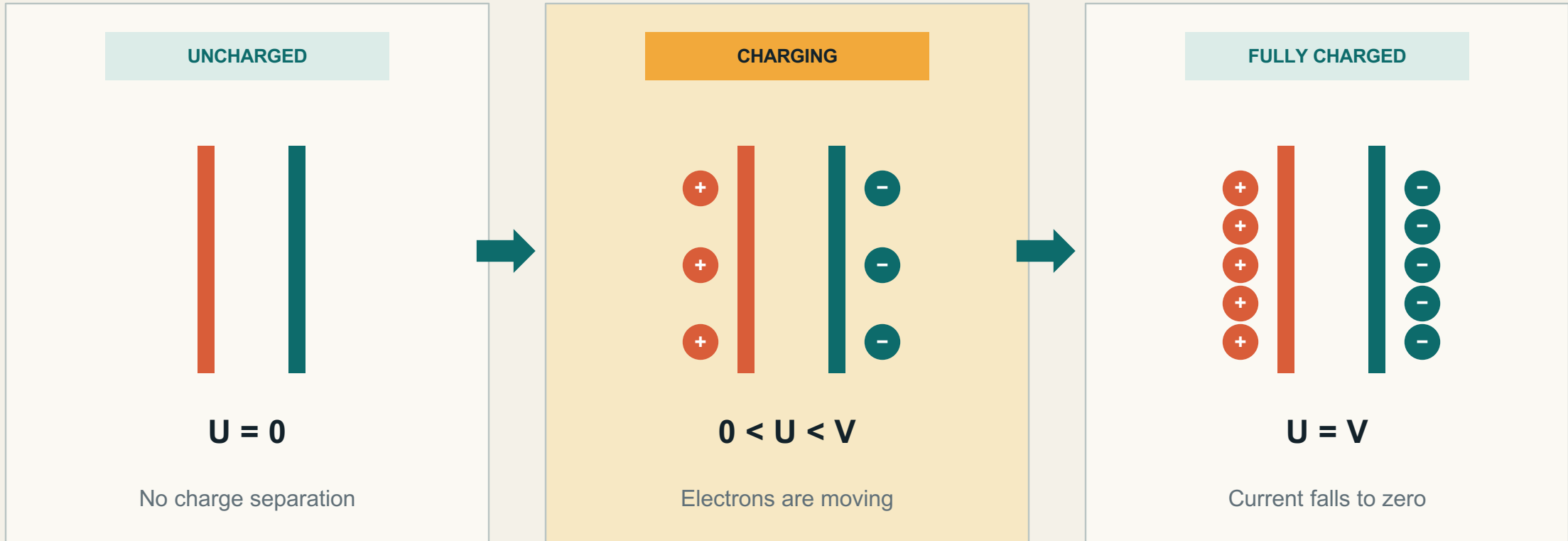
EARLY CHARGE STORAGE



MODERN CAPACITORS

conductors + insulator

A battery moves electrons until the capacitor voltage matches the supply.



More separation → larger potential difference

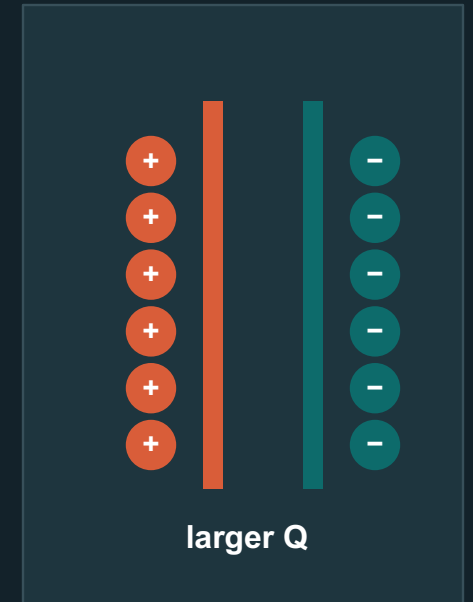
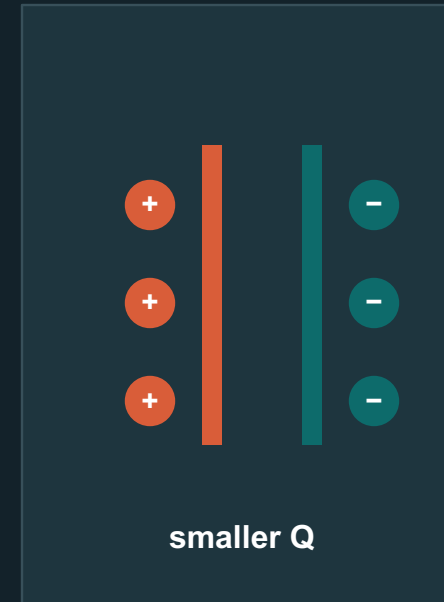
Capacitance is charge stored per volt.

$$C = Q / U$$

Q = magnitude of charge on either plate

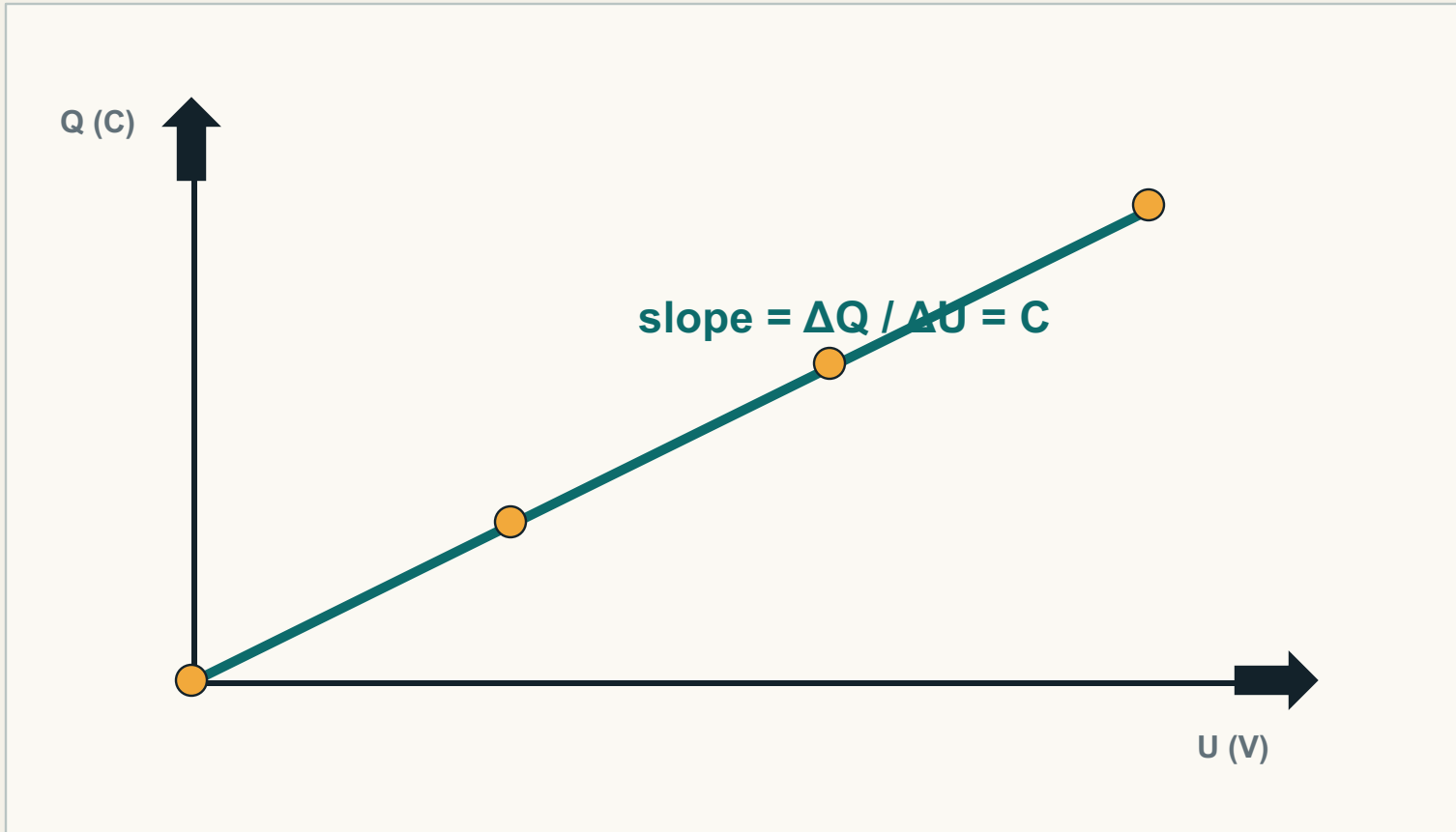
U = potential difference between the plates

Same voltage



Higher capacitance

For one capacitor, $Q \propto U$ —and the slope is C .



Changing Q and U moves the operating point along the same line.

The line's slope stays fixed because C is a property of the capacitor.

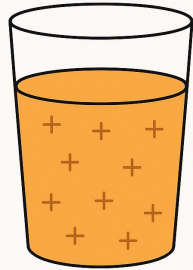
Key distinction

C does not increase just because Q increases.

Water storage is useful—until the analogy starts to mislead.



STORED CHARGE



amount of water ↔ separated charge Q

container capacity ↔ capacitance C

Use the analogy for comparison:

At the same “pressure,” a larger-capacitance device holds more charge.

But remember:

Voltage is potential difference, not literal fill height. Charge is separated across two conductors.

Geometry and dielectric set capacitance.

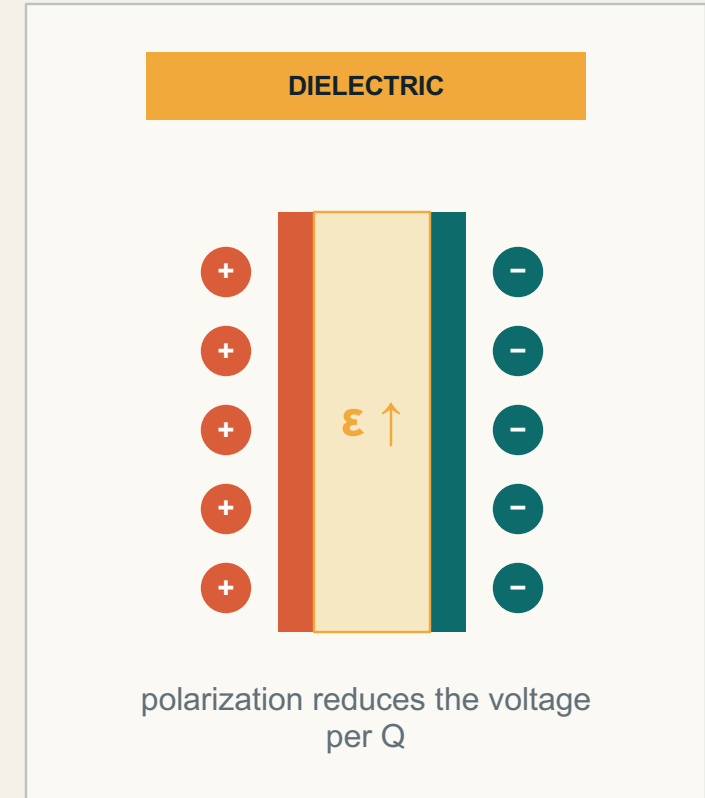
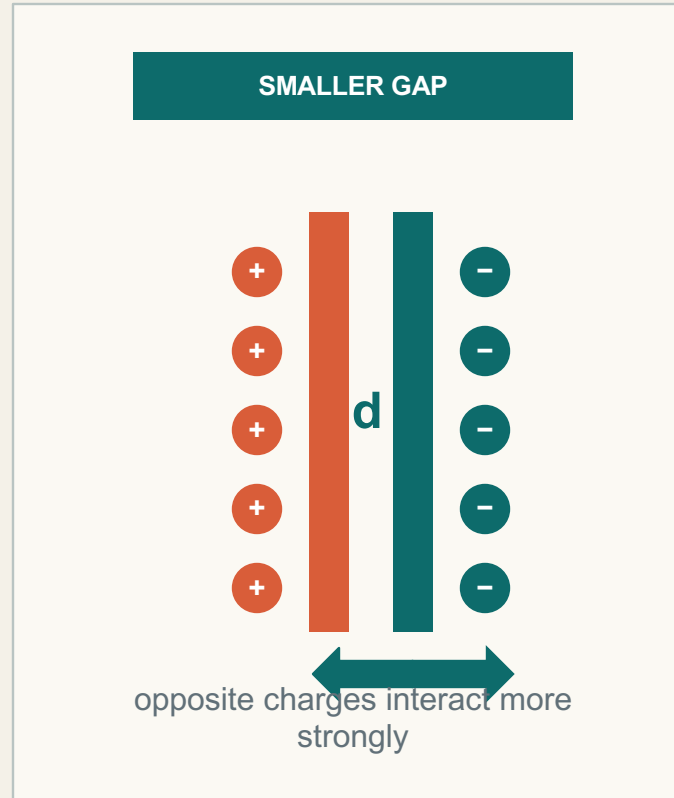
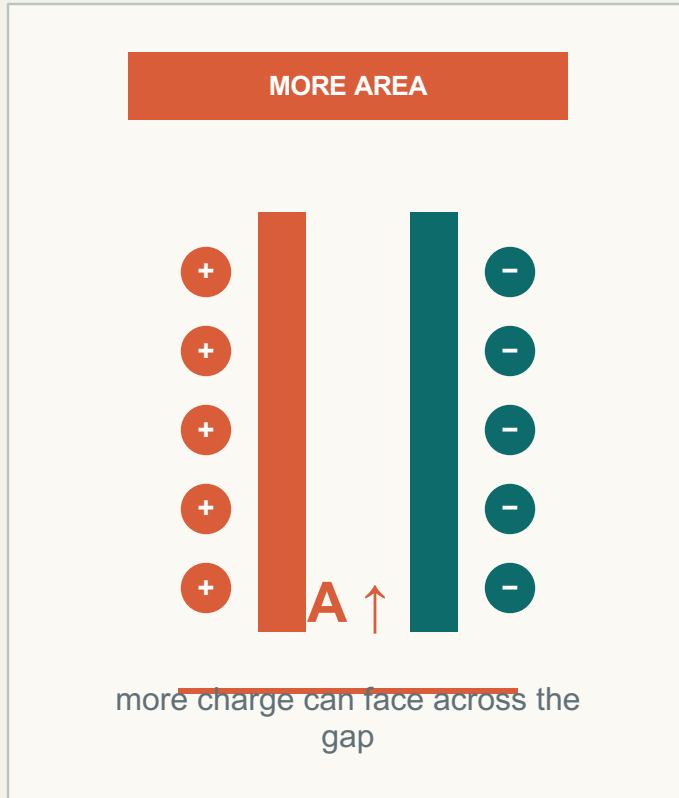
Parallel plates



$$C = \epsilon A / d$$

- A** plate overlap area
larger A → larger C
- d** plate separation
smaller d → larger C
- ε** dielectric response
larger ε → larger C

Three design changes make capacitance larger.



All three changes increase C.

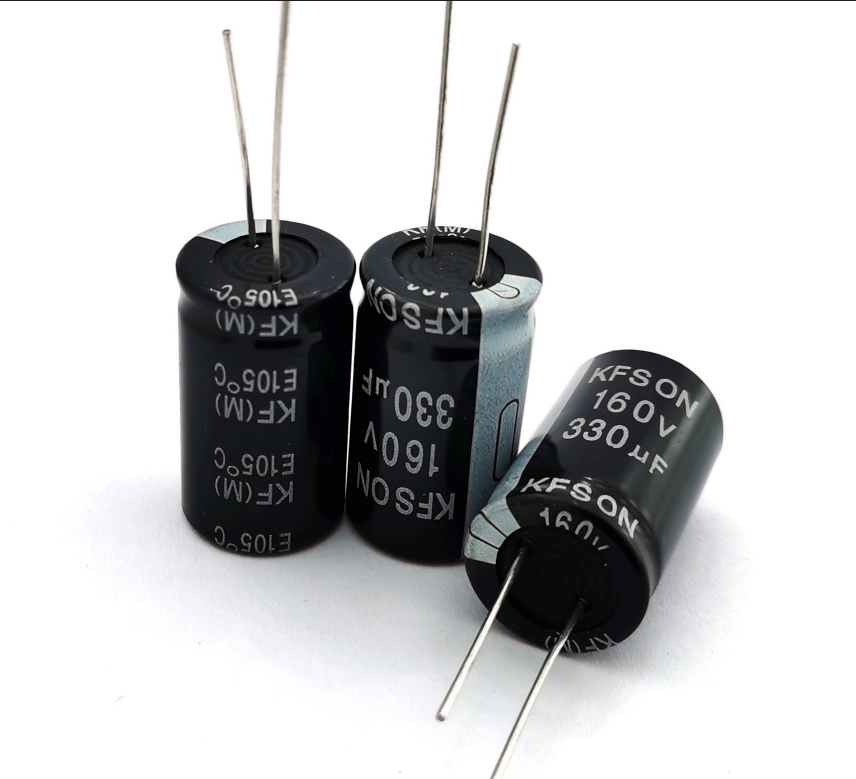
One farad is large, so most components use sub-units.



F		farad 1 F
mF		millifarad 10^{-3} F
µF		microfarad 10^{-6} F
nF		nanofarad 10^{-9} F
pF		picofarad 10^{-12} F

$$1 \text{ F} = 1 \text{ C/V}$$

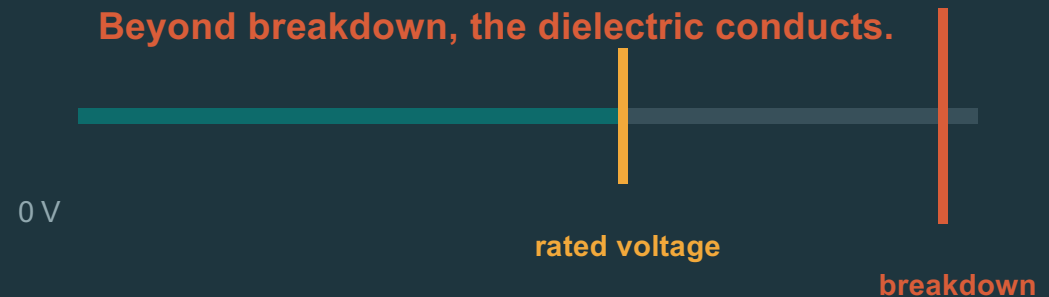
Capacitance is useful only below the voltage limit.



normal operation

Stay at or below the rated voltage.

Beyond breakdown, the dielectric conducts.



breakdown voltage > rated voltage

Use $C = Q/U$ by choosing the unknown first.

A $220\ \mu\text{F}$ capacitor is connected to $12\ \text{V}$. What charge magnitude is stored on either plate?

1

Rearrange

$$Q = CU$$

2

Convert

$$220\ \mu\text{F} = 220 \times 10^{-6}\ \text{F}$$

3

Calculate

$$Q = (220 \times 10^{-6})(12) = 2.64 \times 10^{-3}\ \text{C}$$

ANSWER

2.64 mC

on each plate
(equal magnitude,
opposite sign)

A complete mental model answers three questions.

1

WHAT IS STORED?

Equal-magnitude, opposite charges separated across two conductors.

2

WHAT DOES C MEAN?

Charge stored per volt: $C = Q/U$. For one capacitor, C is constant.

3

WHAT SETS C?

Geometry and dielectric: larger A , smaller d , larger ϵ .

Final check: Why does doubling Q not automatically double C ?